TECHNICAL CASE • STAGE 1

Empório da Música

AI Customer Support Agent

A production-grade, full-stack ReAct and vector RAG architecture

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Live demo: artefact-rovw.onrender.com

65	9	25	6	10	6
products	categories	promotions	agent tools	policy chunks	test scenarios

1. Executive Summary and Problem

Empório da Música is a traditional musical instruments retailer founded in 2008 in Campo Grande/MS. Its support team was overloaded by five high-frequency inquiry types: **catalog pricing, inventory availability, order tracking, physical store logistics** and **consumer return policies**.

I designed and built an autonomous, production-ready AI agent in Python, delivered as a React 19 web app and a Rich terminal CLI. A hybrid orchestration engine combines **Google Gemini 3.1 Flash-Lite** (tool calling), **Qdrant** (vector RAG) and **SQLite / SQLAlchemy** (deterministic relational queries).

Deliverables and constraints met

Requirement	How it was met
Mandatory language	100% Python backend, agent orchestration, database seeding, RAG pipelines and automated test suite.
Full-stack scope	Responsive React 19 frontend with localStorage session persistence, mobile safe-area support and real-time tool-execution feedback.
Live deployment	Running 24/7 on Render with automated CI/CD from GitHub: artefact-rovw.onrender.com.

2. Architecture and Technical Decisions

The core design principle is a clear division of labor: **the LLM interprets intent and writes the answer, while deterministic tools produce every number and every rule it states.**

> **Why hybrid?** Pure RAG fails on arithmetic and inventory, because LLMs hallucinate stock levels and percentages. Pure SQL fails on company policies, because it cannot interpret unstructured legal clauses. The hybrid design sends numbers to SQL and policy text to Qdrant.

Dimension	Technology	Justification
Agent paradigm	Hybrid tool calling (ReAct) + RAG	Deterministic delegation: scalar and transactional data to SQL, unstructured legal and operational text to vector search.
Catalog and orders	SQLite / SQLAlchemy	Strict scalar filtering (price <= 1000), immediate ACID consistency on volatile stock, exact primary-key resolution (Order #17) and exact financial math (5% PIX discount).
Store policies	Qdrant + embeddings	Retrieves unstructured text (políticas_da_loja.pdf), tolerates conversational paraphrasing and grounds answers to prevent legal hallucinations.
LLM provider	Gemini 3.1 Flash-Lite	Fast inference (< 1.5 s), native schema-validated Automatic Function Calling, strong Brazilian Portuguese and stable quotas.
Vector engine	Qdrant (embedded mode)	HNSW indexing. Embedded mode allows zero-dependency local evaluation on disk (qdrant_storage/) with API parity for cloud clusters.
Backend API	FastAPI + Uvicorn	Async ASGI runtime, automatic OpenAPI docs (/docs), strict Pydantic validation and single-container static asset serving.
Frontend	React 19 + Tailwind CSS	Mobile-first chat UI with dynamic viewport height (100dvh), session auto-resume via localStorage and Lucide SVG icons (no text emojis).

2.1 Why relational SQL for products and orders, not embeddings

Embedding catalog and transaction data is a known anti-pattern in commercial AI engineering:

- **Exact numerical inequalities.** Cosine similarity has no notion of numeric boundaries. For "violões até R$ 1.000", vector search can return a R$ 1.299 instrument because the text is similar. SQL enforces `Product.price <= 1000` exactly.
- **Real-time stock volatility.** Stock changes with every checkout and cancellation. Updating vectors means costly re-indexing, while SQL is instantly consistent (Order #18 was correctly detected as out of stock).
- **Deterministic financial arithmetic.** The 5% PIX discount and the 12x installments (minimum R$ 100 per month) are computed in Python from exact SQL decimals, which rules out arithmetic hallucination.
- **Discrete key lookups.** Order IDs (`order_id = 17`) and tracking codes (`BRAB1234567BR`) are exact keys, resolved in O(1).

3. Ingestion Pipelines and Tool Suite

3.1 Relational ingestion (data/*.csv)

Six CSV files were parsed, typed and normalized into relational tables with foreign-key constraints.

Table	Contents
`categories`	9 musical instrument departments
`products`	65 items with prices, specifications and active/discontinued status
`promotions`	25 seasonal campaigns, tracked with an `is_active` flag
`orders`, `order_items`	Tracking codes, fulfillment dates, delivery estimates, cancellation notes
`customers`	Contact information and delivery locations

3.2 Policy chunking and semantic search (políticas_da_loja.pdf)

The internal manual was split into **10 structured chunks** that follow its operational chapters: About Us, Scope Limits, Operating Hours, Payment and Installments, Returns and Remorse, Shipping, Promotions, Warranty and LGPD. Chunks were embedded with `gemini-embedding-001` (3072 dimensions) and stored in Qdrant using Cosine distance.

3.3 Agent tools

The agent calls deterministic Python tools with OpenAPI schema validation:

Tool	Purpose
`get_categories()`	Lists the 9 departments with active product counts.
`query_products(category_name, max_price, min_price, search_text)`	Parametric SQL filtering with automatic 5% PIX discount and active-promotion detection.
`get_product_details(product_name_or_id)`	Technical specifications, current stock and detection of expired promotions.
`check_order_status(order_id, tracking_code, customer_identifier)`	Real-time status, delivery dates, internal cancellation notes and item breakdown.
`get_store_info()`	Store address (Rua 14 de Maio, 3200, Campo Grande/MS), opening hours and contact channels.
`search_store_policies(query)`	Semantic search in Qdrant for business rules, returns and warranties.

4. Persona Engineering and Anti-Hallucination Guardrails

> **Golden Rule (scope).** Empório da Música sells musical instruments only. It does **not** sell accessories such as strings, picks, cables, straps, cases, pedals or standalone amplifiers. The agent acknowledges these requests warmly, explains the limit and redirects the customer to local accessory shops in Campo Grande.

- **Truthfulness policy.** The agent never invents prices, stock or order statuses. Tools are mandatory before any number is stated.
- **Expired promotion transparency.** When a customer mentions an old offer (such as the expired 12% Takamine discount), the agent explains that the campaign ended and presents current prices and standard payment terms.
- **Consumer Defense Code (CDC).** Correctly applies the 7-day remorse window for remote purchases, store-funded reverse logistics and the 10-business-day refund timeline.
- **Zero-emoji policy.** Replies keep a warm, professional, musical tone without text emojis. Icons are rendered only through Lucide React in the UI.

5. Test Scenarios and Transcripts

Six real-world scenarios were run against live data. Full transcripts are in `examples/conversations/`.

#	Customer prompt (pt-BR)	Tool executed	Outcome
1	*Quais opções de violões vocês têm disponíveis custando até R$ 1.000?*	`query_products(category_name='Violões', max_price=1000)`	Returned in-stock guitars (Tagima Memphis AC-39, Rozini RC-104, Shelby SN-100, Giannini GF-1R, Yamaha C40, Giannini GN-15) with table price, 5% PIX price, string type and 12x terms.
2	*Qual o endereço da loja física e quais os horários de funcionamento no sábado e na semana?*	`get_store_info()`	Rua 14 de Maio, 3200, Centro, Campo Grande/MS. Mon–Fri 09:00–18:00, Sat 09:00–13:00, closed Sundays and holidays.
3	*Quanto custa o Takamine GD20? Vi em algum lugar que tinha promoção dele, ainda está com desconto?*	`get_product_details('Takamine GD20')`	R$ 2.199,00 (R$ 2.089,05 on PIX), 5 units in stock. Detected that promotion #4 (12% off) had `is_active=0` and transparently explained it had ended.
4	*Comprei um instrumento pelo WhatsApp faz 4 dias e me arrependi da compra. Posso devolver? Como funciona o frete e o reembolso?*	`search_store_policies('devolução arrependimento prazo frete reembolso')`	Applied Section 4.1: eligible within 7 days of delivery, original packaging and accessories required, reverse shipping fully covered by the store, refund within 10 business days.
5	*Olá! Gostaria de acompanhar a entrega do meu pedido número 1. Qual o status e o código de rastreamento dele?*	`check_order_status(order_id=1)`	Order #1, Maria Silva: Martin D-28 Dreadnought Natural, R$ 11.499,00 via PIX, status Delivered, tracking BRAB1234567BR. Complementary test: Order #18 correctly identified as cancelled for lack of stock.
6	*Vocês vendem encordoamento de aço para violão e palhetas avulsas?*	None (guardrail)	Golden Rule triggered: the agent explained the store's specialization, declined the accessory sale and suggested partner stores in Campo Grande.

6. AI Coding Assistant Workflow

In line with Artefact's evaluation guidelines, this is how AI tools were used during development:

Area	AI contribution	Human contribution
Architecture	None	Defined the hybrid SQL + Qdrant design and the anti-hallucination guardrails.
Data pipeline	Boilerplate SQLAlchemy models for the 6 CSV schemas; PDF parsing routines.	Schema decisions, chunking strategy, validation.
Debugging	Faster diagnosis of strict type deserialization in google-genai (`Union[int, float, None]` for numeric tool arguments).	Root-cause analysis and fix review.
Frontend	Scaffolding of the React 19 + Tailwind component shell.	Stateful chat flows and tool-execution transparency badges.
Testing	Rapid authoring of Pytest fixtures and validation scripts for tools and RAG.	Test strategy and quality gates.

> **Clarification.** The AI assistant was a velocity multiplier for boilerplate, exploratory debugging and test coverage. All architectural choices, business logic, system flows, guardrails and quality gates were decided under direct human direction.

7. Known Limitations and Production Roadmap

Next step	Description
Hybrid lexical search	Combine dense Qdrant embeddings with sparse BM25 via Reciprocal Rank Fusion, to handle alphanumeric product codes.
Multi-agent hierarchy	Split duties into dedicated sub-agents (Sales, Logistics, Policies) orchestrated with LangGraph.
Distributed infrastructure	Move SQLite to managed PostgreSQL with PgBouncer, and local Qdrant to Qdrant Cloud.
Observability and tracing	Integrate OpenInference / Langfuse for token, latency and Ragas faithfulness tracking.